

1 Informative dropout and trial-design choice in
2 aggregated N-of-1 biomarker-validation trials

3 pmsimstats team

4 2026-06-13

5 **Contents**

6	1 Abstract	2
7	2 Introduction	3
8	2.1 A claim about how N-of-1 trials should account for dropout	3
9	2.2 Why the question matters	5
10	2.3 What the published N-of-1 literature does and does not do	8
11	2.4 What this paper contributes	10
12	3 Informative-dropout model	13
13	4 The four design families and their vulnerability windows	16
14	5 Simulation design	20
15	6 Results	22
16	6.1 Power across designs and dropout patterns	22
17	6.2 Bias of the biomarker-treatment interaction estimate	24
18	6.3 Randomization-path decomposition	26
19	6.4 Sensitivity to dropout-pattern misspecification	26
20	7 Design-by-dropout recommendation	28
21	8 Discussion	29
22	8.1 Prototype Monte Carlo confirmation	29
23	9 Conclusions	32
24	10 Conflict of interest	36
25	11 Funding	36
26	12 Data availability statement	36
27	13 Reproducibility	36

29 1 Abstract

30 **Background.** Aggregated N-of-1 trial designs vary substantially in their robust-
 31 ness to informative dropout, yet the published methodological literature has doc-
 32 umented dropout as a generic problem without adequately engaging the coupling
 33 between the trial-design family (open-label, open-label plus blinded discontinu-
 34 ation, traditional crossover, hybrid) and the inferential consequences of biased
 35 dropout for the biomarker-treatment interaction estimand. The unaddressed cou-
 36 pling is not a minor sensitivity issue: the bias and power consequences of dropout
 37 differ by an order of magnitude across design families even at a fixed overall
 38 dropout rate, and the most powerful design under no dropout is not necessarily
 39 the most powerful design under dropout.

40 **Methods.** We formalise the informative-dropout model introduced in¹ (hazard
 41 depending on cumulative symptom worsening since baseline), place it within the
 42 classical missing-data taxonomy, and then reproduce and extend Hendrickson’s
 43 four-design dropout-robustness analysis with wider parameter ranges, sensitiv-
 44 ity to dropout-pattern misspecification at the analysis stage, and an explicit
 45 randomisation-path decomposition. The simulation employs the existing pmsim-
 46 stats framework with extensions to the dropout-pattern grid, and reporting follows
 47 the² ADEMP framework.

48 **Results.** [Forthcoming.] Across all four design families, informative dropout
 49 that selectively retains treatment-non-responders appears to inflate the apparent
 50 biomarker-treatment interaction by underestimating the on-drug response of
 51 the responder stratum. The OL design is the most dropout-robust on power
 52 but contributes no useful biomarker-prediction information; the OL+BDC
 53 and hybrid N-of-1 designs are the most dropout-vulnerable on raw power but,
 54 paradoxically, recover power through the so-called ‘happy accident’ selection
 55 effect (biased dropout preferentially preserves the most-informative crossover
 56 blocks). The traditional crossover design is most robust on combined power-
 57 and-bias under symptom-worsening-driven dropout but most vulnerable under
 58 symptom-improvement-driven dropout, where treatment-responders drop early
 59 and the crossover signal is degraded.

60 **Conclusions.** Trial designers should evaluate proposed N-of-1 designs against
 61 the anticipated dropout-pattern mechanism, not merely against a generic dropout
 62 rate. We provide a design-by-dropout-pattern recommendation table that maps
 63 three classes of dropout mechanism (balanced, worsening-driven, improvement-
 64 driven) to preferred design choices.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[1]

The paper advances two coupled claims about how N-of-1 design selection should account for informative dropout.

- The four design families (OL, OL+BDC, traditional crossover, hybrid) differ in dropout robustness in ways the published literature has not adequately characterised.
- Hendrickson's Figure 4A demonstrates power losses exceeding 50 percent in some design-by-dropout cells while remaining negligible in others.
- The dropout-by-design coupling is a first-order design choice, not a marginal sensitivity issue.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[2]

The first claim is that design families differ in dropout robustness in ways the published literature has not adequately characterised.

- Power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells.
- Design rankings reverse depending on the dropout pattern.
- This is a first-order design choice, not a marginal sensitivity issue.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

104 *The first is that the four candidate trial-design families (open-label, open-*
105 *label plus blinded discontinuation, traditional crossover, and the hybrid N-of-1*
106 *design proposed by¹) differ in their robustness to informative dropout in ways*
107 *that the published methodological literature has not adequately characterised.*
108 *Hendrickson's Figure 4A heatmap demonstrates that the absolute power loss*
109 *from a realistic dropout pattern can exceed 50 percent in some design-by-*
110 *dropout cells while remaining negligible in others, and the design rankings*
111 *reverse depending on the dropout pattern. This is not a marginal sensitivity*
112 *issue but a first-order design choice that the field has not, to the best of our*
113 *knowledge, foregrounded.*

114 [3]

115 **The second claim is that the dropout mechanism, not merely**
116 **the rate, is the methodologically relevant object for design**
117 **selection.**

- 118 • Balanced, worsening-driven, and improvement-driven mecha-
119 nisms interact with design families in qualitatively different
120 ways.
- 121 • Trial designers should evaluate designs against the anticipated
122 dropout mechanism rather than a generic rate.
- 123 • The mechanism is the primary object; the rate is relatively super-
124 ficial.

125 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
126 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
127 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
128 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
129 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
130 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
131 mollit anim id est laborum.

132 *The second claim is that the dropout-pattern mechanism (balanced, symptom-*
133 *worsening-driven, symptom-improvement-driven) interacts with the design*
134 *family in qualitatively different ways than the overall dropout rate alone, and*
135 *that trial designers should therefore evaluate proposed N-of-1 designs against*
136 *the anticipated dropout-pattern mechanism rather than against a generic*
137 *dropout rate. That is, the mechanism is the methodologically relevant ob-*
138 *ject; the rate is relatively superficial.*

139 [4]

140 **The paper imports the mature missing-data taxonomy into**
141 **the N-of-1 design-comparative context for the first time.**

- 142 • MCAR/MAR/MNAR taxonomy and MMRM have been worked
143 through for parallel-group RCTs but not for N-of-1 designs com-
144 paratively.
- 145 • ICH E9 (R1) formalises estimand-aware handling of intercurrent
146 events including dropout.

- 147 • The present paper undertakes the import of this machinery into
148 the N-of-1 design-comparative setting.

149 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
150 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
151 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
152 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
153 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
154 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
155 mollit anim id est laborum.

156 *The case is, in the broadest framing, a special application of the missing-*
157 *data taxonomy of Little and Rubin³ to the N-of-1 design class. The taxon-*
158 *omy is mature: missing-completely-at-random (MCAR), missing-at-random*
159 *(MAR), and missing-not-at-random (MNAR) define a ladder of assumptions*
160 *about the missingness mechanism, and each layer imposes correspondingly*
161 *stronger inferential conditions^{4,5}. The classical clinical-trial literature has*
162 *worked through the implications for parallel-group randomised trials^{6–8} and*
163 *has converged on the mixed-effects model for repeated measures (MMRM) as*
164 *the default analytic strategy when dropout is plausibly MAR. The ICH E9*
165 *(R1) addendum⁹ formalises the estimand-aware framework that handles in-*
166 *tercurrent events including dropout¹⁰. Unfortunately, none of this machinery*
167 *has been imported into N-of-1 methodology in a design-comparative way. The*
168 *present paper undertakes that import.*

169 2.2 Why the question matters

170 [5]

171 **Three reasons motivate the design-by-dropout coupling**
172 **specifically for the aggregated N-of-1 design class.**

- 173 • Informative dropout can selectively destroy within-participant
174 contrasts central to the N-of-1 estimand.
- 175 • Patient self-selection is correlated with latent biomarker-response
176 components that are partly signal, not nuisance.
- 177 • The four canonical designs differ in their phase-specific vulnera-
178 bility windows.

179 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
180 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
181 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
182 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
183 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
184 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
185 mollit anim id est laborum.

186 *Three reasons motivate the design-by-dropout coupling specifically for the*
187 *aggregated N-of-1 design class. We shall take each in turn.*

188 [6]

189
190

191
192
193
194
195
196

197
198
199
200
201
202
203

204
205
206
207
208
209
210
211
212
213
214
215
216

217

218
219

220
221
222
223
224
225
226
227

228
229
230
231

The first reason is that informative dropout selectively destroys within-participant contrasts, not merely sample size.

- Unlike parallel-group RCTs, N-of-1 dropout removes specific within-participant contrasts central to the estimand.
- A participant lost before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation.
- A 30-percent phase-selective dropout can degrade interaction power more than 50-percent uniform dropout.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The first reason is that the N-of-1 estimand depends on within-participant contrasts that informative dropout can selectively destroy. In a parallel-group RCT, dropout reduces the effective sample size but does not, under MAR, systematically distort the per-arm mean response that the analysis estimates. In an aggregated N-of-1 trial, dropout removes specific within-participant contrasts: a participant who drops out after the open-label phase but before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation that the design is constructed to deliver. The information loss is not merely in the sample size but in the combinatorial structure of the design. A 30-percent overall dropout rate that selectively removes participants from the post-discontinuation phase can degrade the biomarker-treatment interaction power more than a 50-percent dropout rate that drops uniformly across phases.

[7]

The second reason is that patient self-selection is correlated with latent components that are partly signal, not nuisance.

- High-BR patients tend to remain on drug; high-PB patients may drop at discontinuation; negative-TV patients may drop from worsening.
- Each mechanism produces a different dropout pattern that interacts differently with the design family.
- The worsening- and improvement-driven patterns operationalise the responder-versus-non-responder distinction the biomarker literature is validating.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in

232 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
233 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
234 mollit anim id est laborum.

235 *The second reason is that patient self-selection into and out of the trial is cor-*
236 *related with the latent BR-PB-TV components in ways that are not nuisance*
237 *but partly signal. Patients with strong pharmacological response (high BR)*
238 *are likely to remain on drug and therefore to remain in the trial; patients with*
239 *strong placebo response (high PB) may drop out at the blinded-discontinuation*
240 *phase when their symptoms return on the silently-switched placebo, mistaking*
241 *the return for trial failure; patients with adverse natural-history trajectories*
242 *(negative TV) may drop out from worsening regardless of treatment. Each*
243 *of these mechanisms produces a different dropout pattern, and each interacts*
244 *with the design family in a different way. The symptom-worsening-driven'*
245 *andsymptom-improvement-driven' patterns of¹ are operationalisations of the*
246 *responder-versus- non-responder distinction that the predictive-biomarker lit-*
247 *erature is endeavoring to validate; the dropout pattern itself is therefore not*
248 *incidental.*

249 [8]

250 **The third reason is that each design family has distinct phase-**
251 **specific vulnerability windows whose trade-offs this paper**
252 **analyses.**

- 253 • OL is most dropout-robust but cannot identify the BR-versus-PB
254 split required for the predictive-biomarker question.
- 255 • OL+BDC introduces a discontinuation-phase window where re-
256 sponders may drop from symptom return.
- 257 • The hybrid design has the largest vulnerability surface; the trade-
258 off between identifiability and dropout vulnerability is the central
259 design choice.

260 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
261 mod tempor incidunt ut labore et dolore magna aliqua. Ut enim ad
262 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
263 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
264 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
265 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
266 mollit anim id est laborum.

267 *The third reason is that the four canonical designs differ in their vulnerability*
268 *windows. The open-label design has no discontinuation- driven dropout and*
269 *is therefore most robust to dropout overall, but it cannot identify the BR-*
270 *versus-PB split and is the wrong design for the predictive-biomarker ques-*
271 *tion. The OL+BDC design supplies identifiability of the split but introduces a*
272 *discontinuation-phase vulnerability window where treatment-responders may*
273 *drop out from symptom return. The traditional crossover design has two dis-*
274 *continuation windows (one per crossover) and is more exposed to symptom-*
275 *improvement-driven dropout. The hybrid N-of-1 design has the largest design*
276 *surface area and therefore the largest set of vulnerability windows. The trade-*

277 *off between identifiability and dropout vulnerability is the central design choice*
278 *this paper analyses.*

279 2.3 What the published N-of-1 literature does and does not do

280 [9]

281 **The methodological N-of-1 corpus addresses dropout gener-**
282 **ically but does not engage the design-by-dropout coupling**
283 **this paper centres on.**

- 284 • Lillie et al. and Vohra et al. CONSORT extension codify report-
- 285 ing standards but do not prescribe analytic methods for informa-
- 286 tive dropout in N-of-1 contexts.
- 287 • Systematic reviews document that dropout rates are reported
- 288 without specifying mechanism or analytic treatment.
- 289 • Dropout is one of the leading methodological weaknesses in N-of-1
- 290 reporting audits.

291 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-

292 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

293 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip

294 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in

295 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur

296 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt

297 mollit anim id est laborum.

298 *The methodological N-of-1 corpus addresses dropout as a generic challenge*
299 *but does not engage the design-by-dropout coupling this paper centres on. We*
300 *briefly survey what has been broached and what has not.*

301 [10]

302 **Existing N-of-1 reporting standards codify dropout disclo-**
303 **sure but do not prescribe analytic methods for informative**
304 **dropout.**

- 305 • Lillie et al. and the Vohra CONSORT extension set reporting
- 306 standards without addressing informative-dropout analysis.
- 307 • Systematic reviews find that substantial proportions of N-of-1
- 308 studies report rates without specifying mechanism or analytic
- 309 treatment.
- 310 • Dropout is one of the leading methodological weaknesses in N-of-1
- 311 reporting audits.

312 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-

313 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

314 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip

315 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in

316 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur

317 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
318 mollit anim id est laborum.

319 *Lillie et al.¹¹ and the CONSORT extension of Vohra et al.¹² codify reporting*
320 *standards that include dropout reporting but do not prescribe analytic methods*
321 *for handling informative dropout specifically in N-of-1 contexts. The system-*
322 *atic reviews of Stunnenberg et al.¹³ and Natesan Batley and colleagues¹⁴ doc-*
323 *ument that substantial proportions of published N-of-1 studies report dropout*
324 *rates without specifying the dropout mechanism or its analytic treatment, and*
325 *that dropout is one of the leading methodological weaknesses identified in*
326 *N-of-1 reporting audits.*

327 [11]

328 **Hendrickson (2020) is the foundational reference; this paper**
329 **extends it along three axes.**

- 330 • The extension uses wider parameter ranges (carryover half-lives,
331 effect sizes, dropout rates) than the original.
- 332 • It examines sensitivity to dropout-pattern misspecification at the
333 analysis stage, which the original assumed away.
- 334 • It offers an explicit decomposition of the happy-accident
335 randomisation-path selection effect, scoped for path-stratified
336 production testing.

337 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
338 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
339 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
340 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
341 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
342 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
343 mollit anim id est laborum.

344 ¹ introduce the first quantitative analysis of dropout- by-design coupling in the
345 aggregated N-of-1 context, with a hazard- based dropout model whose intensity
346 depends on cumulative symptom worsening since baseline. Their Figure 4A
347 shows the four-design power matrix as a function of dropout pattern, and their
348 Section 3.4 bias- quantification establishes that under symptom-worsening-
349 driven dropout the OL+BDC design develops a substantial negative bias in the
350 biomarker-treatment interaction coefficient ($p < 0.0001$). The present paper
351 takes¹ as the foundational reference and extends it along three axes. First,
352 we use wider parameter ranges (longer carryover half-lives, higher biomarker
353 effect sizes, larger dropout rates) than the original analysis. Second, we
354 examine sensitivity to dropout-pattern misspecification at the analysis stage:¹
355 assumed the analyst knows the dropout mechanism, but in practice the analyst
356 may model dropout as MCAR while the truth is MNAR, or may model it as
357 symptom-worsening-driven when it is improvement-driven, and the present
358 paper quantifies the inferential cost of each misspecification. Third, we offer
359 an explicit decomposition of the ‘happy accident’ randomisation-path selection
360 effect.¹ noted that biased dropout preferentially preserves the most-informative
361 crossover blocks in OL+BDC and hybrid N-of-1 designs (because patients

with inconclusive open-label response are more likely to continue to the later blocks); in the present prototype we offer a marginal account of this effect, and a path-stratified test is scoped for the production rerun.

[12]

The paper’s contribution is to import classical missing-data, joint-modelling, and selection-bias machinery into the N-of-1 design-comparative simulation context.

- Little-Rubin taxonomy, selection models, and pattern-mixture models are adapted to the N-of-1 setting.
- Hazard-coupled-to-longitudinal-trajectory joint-modelling underwrites the Hendrickson dropout model and its extension.
- None of this machinery has previously appeared in the N-of-1 literature in a design-comparative simulation study.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The classical missing-data literature^{3,4,15} supplies the methodological scaffolding (MAR/MNAR taxonomy, selection models, pattern-mixture models, multiple imputation) that we adapt to the N-of-1 setting. The informative-dropout and joint-modelling machinery^{16–18} supplies the hazard-coupled-to-longitudinal-trajectory formulation that underwrites the¹ dropout model and its extension here, and the longitudinal mixed-models foundations¹⁹ supply the analytic backbone for the $\beta_{bm:D}$ inference under MAR-plausible regimes. The clinical-trial dropout literature^{6–8} supplies the parallel-group analogue against which the design-comparative results below should be read, and the selection-bias literature²⁰ supplies the template for thinking about how response-based exclusion or attrition damages internal and external validity. None of this machinery has been imported into the N-of-1 methodology literature in a design-comparative simulation study, and that import is the present paper’s contribution.

2.4 What this paper contributes

[13]

The paper makes four contributions across Sections 2 through 6.

- Section 2 formalises the informative-dropout model and places it in the missing-data taxonomy.
- Section 3 characterises each design family’s phase-specific vulnerability windows.

- 404 • Sections 4 and 5 conduct and report a Monte Carlo simulation
405 crossing four designs against five dropout patterns.

406 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
407 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
408 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
409 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
410 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
411 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
412 mollit anim id est laborum.

413 *We make four contributions, each addressed in a section below.*

414 [14]

415 **Section 2 formalises the hazard-based informative-dropout**
416 **model and establishes its MNAR character explicitly.**

- 417 • The Hendrickson hazard model is placed in the MCAR/MAR/MNAR
418 taxonomy and shown to be MNAR.
- 419 • Four canonical dropout patterns plus a no-dropout reference are
420 defined as cells in the (β_0, β_1) hazard-parameter space.
- 421 • The operational definitions fix the simulation’s data-generating
422 mechanism for the subsequent studies.

423 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
424 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
425 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
426 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
427 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
428 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
429 mollit anim id est laborum.

430 *In Section 2 we formalise the informative-dropout model for aggregated N-of-*
431 *1 trials. The hazard-based model of^l is placed in the missing-data taxonomy,*
432 *its MNAR character is established explicitly, and the four canonical dropout*
433 *patterns (balanced, more-of-flat, more- of-biased, high-dropout, plus a fifth no-*
434 *dropout reference) are defined operationally as parameter cells in the (β_0, β_1)*
435 *space of the hazard.*

436 [15]

437 **Section 3 identifies the phase-specific dropout-vulnerability**
438 **windows and at-risk within-participant contrasts for each de-**
439 **sign family.**

- 440 • Each of the four design families is characterised by its exposure
441 to each dropout pattern per phase.
- 442 • Within-participant contrasts that are at risk of removal under
443 each pattern are identified explicitly.

- 444 • The characterisation motivates the design-by-dropout-pattern
445 structure of the simulation in Sections 4 and 5.

446 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
447 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
448 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
449 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
450 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
451 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
452 mollit anim id est laborum.

453 *In Section 3 we characterise the four design families' dropout vulnerability*
454 *windows. For each of the open-label, OL+BDC, traditional crossover, and*
455 *hybrid N-of-1 designs we identify the phase-specific exposure to each dropout*
456 *pattern and the within-participant contrasts that are at risk of being removed.*

457 [16]

458 **Sections 4 and 5 report a Monte Carlo study crossing four**
459 **designs against five dropout patterns at three sample sizes**
460 **and three effect sizes.**

- 461 • Performance measures include power, type I error, bias, empirical
462 SE, and path-conditional power decomposition.
- 463 • Reporting follows the Morris (2019) ADEMP framework with
464 Monte Carlo standard errors alongside every estimate.
- 465 • The randomisation-path decomposition quantifies the happy-
466 accident selection effect.

467 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
468 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
469 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
470 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
471 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
472 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
473 mollit anim id est laborum.

474 *In Sections 4 and 5 we conduct and report a Monte Carlo simulation study*
475 *that crosses the four design families against the five dropout patterns at three*
476 *sample sizes ($N \in \{35, 70, 100\}$) and three biomarker-effect sizes ($c_{bm} \in$*
477 *$\{0, 0.3, 0.6\}$). The simulation reports power, type I error, bias of $\hat{\beta}_{bm:D}$, mean*
478 *standard error, and the randomisation-path-conditional power decomposition*
479 *that quantifies the 'happy accident' selection effect. Reporting follows the²*
480 *ADEMP framework adopted as the project-wide simulation-reporting stan-*
481 *dard.*

482 [17]

483 **Section 6 synthesises a recommendation table mapping**
484 **dropout mechanism class to preferred N-of-1 design choice.**

- 485 • Three mechanism classes (balanced, worsening-driven,

improvement-driven) are mapped to preferred designs indexed by anticipated dropout rate.

- The table is presented as practical guidance for trial designers selecting among the four canonical N-of-1 families.
- Section 7 discusses implications for trial-design selection and pmsimstats operating characteristics.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

In Section 6 we synthesise a design-by-dropout-pattern recommendation table that maps three classes of dropout mechanism (balanced, worsening- driven, improvement-driven) to preferred design choices indexed by the anticipated overall dropout rate. The recommendation is presented as guidance for trial designers selecting among the four canonical N-of-1 design families when a particular dropout mechanism is anticipated.

[18]

Section 7 discusses implications for trial-design selection and pmsimstats operating characteristics.

- The discussion addresses design selection for biomarker-validation N-of-1 trials.
- Operating characteristics of the pmsimstats simulation programme are reviewed.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Section 7 discusses the implications for trial-design selection in biomarker-validation N-of-1 trials and for the operating characteristics of the pmsimstats simulation programme.

3 Informative-dropout model

[19]

The censoring hazard combines a time-only non-informative component with a symptom-trajectory-biased component,

made absorbing by a monotone carry-forward pass.

- The per-timepoint censoring probability is the union of two independent draws: a flat component scaling with elapsed time and a biased component scaling with the convex-transformed symptom change since baseline.
- Two hazard parameters govern the split: β_0 sets the overall dropout fraction and β_1 the proportion of that fraction routed through the symptom-biased component, with the convexity exponent e_{b1} fixed at two.
- A second carry-forward pass renders dropout absorbing: once a participant is censored at a timepoint, all later timepoints for that participant are censored.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

We adopt the hazard-based informative-dropout model implemented in the pmsimstats R/censordata.R infrastructure, which operationalises the mechanism of^l. For participant i at timepoint t , the marginal censoring probability is the union (logical OR) of two independent Bernoulli draws,

$$\Pr(\text{drop}_{it}) = 1 - (1 - p_{it}^{\text{flat}})(1 - p_{it}^{\text{bias}}),$$

where the flat component $p_{it}^{\text{flat}} \propto t$ depends on elapsed time alone and is calibrated to a marginal dropout fraction of $\beta_0(1 - \beta_1)$, and the biased component $p_{it}^{\text{bias}} \propto t(\Delta_{Sx}(t) + 100)^{e_{b1}}$ depends on the symptom change $\Delta_{Sx}(t)$ since baseline and is calibrated to $\beta_0\beta_1$. The additive shift of 100 keeps the convex argument positive across the realised range of symptom scores, so that a large improvement lowers the dropout probability while a small change or a worsening raises it. A second pass enforces monotone dropout: once a participant is censored at a timepoint, every later timepoint for that participant is also set missing. The hazard is thus governed by three quantities: the overall rate β_0 , the biased proportion β_1 , and the convexity exponent e_{b1} , held at two throughout.

[20]

Whenever the biased component is active the mechanism is missing-not-at-random, because the censoring indicator at a timepoint depends on the symptom value that the same indicator renders missing.

- The flat component depends only on elapsed time, a fully observed design variable, and is therefore non-informative about the latent response.

- The biased component makes dropout at time t depend on $\Delta_{Sx}(t)$, and because that observation is itself censored at the triggering timepoint the mechanism is MNAR in the Little-Rubin sense.
- The departure from ignorability is controlled by β_1 : at $\beta_1 = 0$ the mechanism reduces to the non-informative limit, and the MNAR departure increases monotonically with β_1 .

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The two components occupy different rungs of the Little-Rubin ladder. The flat component depends only on elapsed time, which is a fully observed design variable rather than a function of the response, so on its own it induces missingness that is ignorable for likelihood-based inference. The biased component is the consequential one. Because the censoring indicator for the observation at time t is a function of $\Delta_{Sx}(t)$, the symptom change realised at that same timepoint, and because the triggering observation is itself rendered missing in the carry-forward pass, the missingness of Y_{it} depends on the value of Y_{it} . This is the defining condition for missing-not-at-random data^{3,4}. The mechanism is therefore MNAR for every pattern with $\beta_1 > 0$, and the standard MMRM appeal to a MAR justification⁶ does not hold. The extent of the violation scales with β_1 , which makes the biased proportion, not the overall rate, the methodologically load-bearing parameter.

[21]

Five dropout patterns are fixed as cells in the (β_0, β_1) hazard-parameter space, spanning a no-dropout reference through balanced, flat-weighted, bias-weighted, and high-rate conditions.

- The no-dropout reference sets $\beta_0 = 0$ and supplies the full-data gold standard against which the bias estimands of Study 2 are measured.
- Balanced ($\beta_0 = 0.05, \beta_1 = 0.5$), more-of-flat ($\beta_1 = 0.2$), and more-of-biased ($\beta_1 = 0.8$) hold the overall rate fixed while varying informativeness from mostly-ignorable to strongly MNAR.
- High-dropout ($\beta_0 = 0.15, \beta_1 = 0.5$) raises the overall rate at the balanced split, isolating the rate axis from the informativeness axis.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in

612 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
613 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
614 mollit anim id est laborum.

615 *The five dropout patterns are defined operationally as cells in the (β_0, β_1)*
616 *parameter space of the hazard, with $e_{b1} = 2$ held throughout. The no-dropout*
617 *reference ($\beta_0 = 0$) retains the complete data and furnishes the gold-standard*
618 *estimate of $\beta_{bm:D}$ used in the second bias-reference mode of Study 2. The*
619 *balanced pattern ($\beta_0 = 0.05, \beta_1 = 0.5$) routes half of a modest five-percent*
620 *dropout fraction through the symptom-biased component. Holding β_0 fixed,*
621 *the more-of-flat ($\beta_1 = 0.2$) and more-of-biased ($\beta_1 = 0.8$) patterns move the*
622 *same overall rate toward the ignorable and the strongly MNAR ends of the in-*
623 *formativeness axis respectively, so that any difference in operating character-*
624 *istics between them isolates the effect of the mechanism from the effect of the*
625 *rate. The high-dropout pattern ($\beta_0 = 0.15, \beta_1 = 0.5$) triples the overall rate*
626 *at the balanced split, isolating the complementary rate axis. The sign conven-*
627 *tion of the biased component is worsening-driven, in that a worsening relative*
628 *to baseline raises the dropout probability; the symptom-improvement-driven*
629 *variant examined in the design-recommendation of Section 6 is obtained by*
630 *reflecting the symptom-change argument, so that improvement rather than*
631 *worsening raises the hazard.*

632 4 The four design families and their vulnerability win- 633 dows

634 [22]

635 **The four design families are distinguished by their phase**
636 **structure, which determines both the within-participant con-**
637 **trasts that identify the biomarker-treatment interaction and**
638 **the windows in which informative dropout can remove them.**

- 639 • Each design is a set of randomisation paths through an ordered
640 sequence of on-drug and off-drug phases, encoded by the **ondrug**
641 path vectors of **buildtrialdesign**.
- 642 • The estimand-defining contrasts are within-participant dif-
643 ferences across phases; informative dropout is damaging in
644 proportion to how directly it removes those contrasts rather
645 than merely reducing sample size.
- 646 • The four designs are ordered by increasing design-surface area,
647 and therefore by an increasing number of vulnerability windows,
648 from open-label through the hybrid N-of-1 design.

649 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
650 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
651 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
652 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
653 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
654 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt

655 mollit anim id est laborum.

656 *Each candidate design is represented as a set of randomisation paths through*
657 *an ordered sequence of on-drug and off-drug phases, encoded by the `on drug`*
658 *path vectors that `buildtrialdesign` accepts. The biomarker-treatment in-*
659 *teraction is identified by within-participant contrasts across those phases, so*
660 *the consequence of informative dropout for a given design is determined by*
661 *which phase a dropout removes and which contrast that phase carries. We*
662 *take the four designs in order of increasing design-surface area.*

663 [23]

664 **The open-label design has a single all-on-drug path, no dis-**
665 **continuation window, and is the most dropout-robust but**
666 **cannot identify the pharmacological-versus-placebo split.**

- 667 • The single path carries only an on-drug-versus-baseline contrast,
668 which is confounded with natural history and cannot separate
669 the *BR* and *PB* components.
- 670 • With no off-drug phase there is no discontinuation-driven vulner-
671 ability window; dropout reduces only the precision of the on-drug
672 slope.
- 673 • The design is the robustness benchmark but is the wrong design
674 for the predictive-biomarker question.

675 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
676 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
677 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
678 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
679 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
680 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
681 mollit anim id est laborum.

682 *The open-label design comprises a single path on which every timepoint is on-*
683 *drug. The only within-participant contrast available is the on-drug response*
684 *against baseline, which is confounded with the natural-history trajectory *TV**
685 *and cannot separate the pharmacological response *BR* from the placebo re-*
686 *sponse *PB*. Because there is no off-drug phase, there is no discontinuation-*
687 *driven vulnerability window, and informative dropout can do no more than*
688 *reduce the precision with which the on-drug slope is estimated. The design is*
689 *consequently the most robust to dropout of the four and serves as the robust-*
690 *ness benchmark, but its inability to identify the *BR*-versus-*PB* split makes*
691 *it the wrong instrument for the predictive-biomarker question that motivates*
692 *the programme.*

693 [24]

694 **The open-label-plus-blinded-discontinuation design adds a**
695 **silent-placebo phase that identifies the pharmacological-**
696 **versus-placebo split but opens a discontinuation window**
697 **in which worsening-driven dropout removes exactly that**
698 **contrast.**

- The estimand-defining contrast is the on-drug-versus-post-discontinuation response, which separates BR from PB .
- Treatment-responders whose symptoms return on the silently switched placebo may drop, mistaking the return for trial failure, which is a worsening-driven dropout located in the discontinuation phase.
- Because the lost phase carries the identifying contrast, dropout here biases $\hat{\beta}_{bm:D}$ rather than merely inflating its variance.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The open-label-plus-blinded-discontinuation design retains the open-label phase and appends a blinded discontinuation phase in which the active drug is silently replaced by placebo. The discontinuation phase supplies the identifiability that the bare open-label design lacks: the contrast between the on-drug response and the post-discontinuation response separates the pharmacological component BR from the placebo component PB . That same phase is the design's vulnerability window. A treatment-responder whose symptoms return on the silently switched placebo may interpret the return as trial failure and drop out, a worsening-driven dropout concentrated in the discontinuation phase. The damage is more than a loss of precision: because the removed phase carries the very contrast that defines the estimand, dropout here produces a systematic negative bias in $\hat{\beta}_{bm:D}$, the effect quantified in¹ Section 3.4 and reproduced in Study 2.

[25]

The traditional crossover design has two randomisation paths and two transition windows, and is robust to worsening-driven dropout but exposed to improvement-driven dropout at each crossover.

- The drug-first and placebo-first paths each carry a drug-block-versus-placebo-block contrast that identifies the within-participant treatment effect.
- A responder who feels well on the drug block and worsens after crossing to placebo may drop, an improvement-driven dropout that removes the second block and one crossover contrast.
- The two transition windows make the design more exposed than the single-window OL+BDC under improvement-driven dropout, though comparably robust under worsening-driven dropout.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

743 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
744 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
745 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
746 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
747 mollit anim id est laborum.

748 *The traditional crossover design has two randomisation paths, one beginning*
749 *on drug and crossing to placebo and the other the reverse, each composed of*
750 *two blocks separated by a transition. The identifying contrasts are the drug-*
751 *block-versus-placebo-block differences carried within each path. The design has*
752 *two transition windows, one per crossover, and its exposure is asymmetric in*
753 *the dropout mechanism. Under worsening-driven dropout it is comparatively*
754 *robust, because worsening tends to occur on the placebo block and a dropout*
755 *there still leaves the drug-block observations that anchor the contrast. Under*
756 *improvement-driven dropout it is exposed: a responder who feels well on the*
757 *drug block and then worsens after crossing to placebo may drop, removing the*
758 *second block and with it one of the two crossover contrasts. The two transition*
759 *windows therefore make the crossover design more vulnerable than the single-*
760 *window OL+BDC design when the operative mechanism is improvement-*
761 *driven.*

762 [26]

763 **The hybrid N-of-1 design has the largest design surface and**
764 **the most vulnerability windows, but recovers power through**
765 **the happy-accident selection effect that preferentially pre-**
766 **serves the most informative crossover blocks.**

- 767 • An open-label run-in followed by a randomised multi-block
768 crossover gives four randomisation paths and the widest set of
769 vulnerability windows.
- 770 • Participants with a clear open-label response are resolved early,
771 so those continuing to the later randomised blocks are enriched
772 for inconclusive open-label response.
- 773 • Biased dropout therefore preferentially preserves the most infor-
774 mative crossover blocks, partially offsetting the raw power loss
775 from the larger vulnerability surface.

776 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
777 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
778 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
779 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
780 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
781 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
782 mollit anim id est laborum.

783 *The hybrid N-of-1 design of^l combines an open-label run-in, which estab-*
784 *lishes each participant's apparent response, with a subsequent randomised*
785 *multi-block crossover. With two randomisation points it admits four paths*
786 *and the largest design-surface area of the four families, and therefore the*
787 *widest set of phase-specific vulnerability windows: the run-in phase, each*

crossover block, and each transition between them. On raw power this makes the hybrid design the most dropout-exposed. The exposure is partly offset, however, by a selection effect that¹ terms a ‘happy accident’. Participants whose open-label response is unambiguous, whether clear responders or clear non-responders, are the ones most readily resolved and most prone to leave early; those who continue into the later randomised blocks are consequently enriched for inconclusive open-label response, which is precisely the subgroup whose crossover contrasts are most informative for the biomarker-treatment interaction. Biased dropout thus preferentially preserves the most informative crossover blocks, partially recovering the power that the larger vulnerability surface would otherwise forfeit. The magnitude of this recovery is the object of the randomisation-path decomposition reported in Study 3.

5 Simulation design

[27]

The simulation follows ADEMP with power as primary estimand and four secondary estimands covering bias, empirical SE, path decomposition, and misspecification sensitivity.

- Primary estimand is power for the $H_0 : \beta_{bm:D} = 0$ interaction test in the post-dropout linear-mixed analysis.
- Bias is reported in two modes: relative to the calibrated true value and relative to the full-data gold standard.
- Secondary estimands include path-conditional power (Study 3) and dropout-pattern misspecification sensitivity (Study 4).

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The simulation programme follows the ADEMP framework of². The primary estimand is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear-mixed analysis fitted to the post-dropout retained data. Secondary estimands are the bias of $\hat{\beta}_{bm:D}$ (reported in two reference modes: relative to the calibrated true value and relative to the gold-standard estimate from the full uncensored data), the empirical Monte Carlo SE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decomposition that quantifies the ‘happy accident’ selection effect (Study 3), and the analysis-method sensitivity to dropout-pattern misspecification (Study 4).

[28]

Every performance measure is reported with its Monte Carlo standard error, following the project-wide ADEMP-audit recommendation.

- Power and type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- Bias and empirical SE of $\hat{\beta}_{bm:D}$ carry the standard Monte Carlo SE of the within-replicate mean.
- Reporting MCSEs alongside every estimate is a project-wide ADEMP-audit requirement.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Performance measures are reported with Monte Carlo standard errors alongside every estimate, addressing the project-wide ADEMP-audit recommendation. Power and type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$; bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean.

[29]

Four pre-registered studies fix the simulation scope before production runs: power, bias, path decomposition, and misspecification sensitivity.

- Study 1 (180 cells) reproduces and extends Hendrickson Figure 4A for power as a function of design and dropout pattern.
- Study 2 (120 cells) addresses two-mode bias quantification; Study 3 (12 cells) examines the happy-accident path decomposition.
- Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at [analysis/scripts/informative-dropout-by-design/00-ademp-pre-reg](#) which fixes four studies before production runs. Study 1 (180 cells) covers power as a function of design and dropout pattern, reproducing and extending¹ Figure 4A. Study 2 (120 cells) addresses two-mode bias quantification. Study 3 (12 cells with path-conditional decomposition) examines the ‘happy accident’ randomisation-path selection effect. Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

(Forthcoming detail. Pmsimstats infrastructure with the existing R/censordata.R hazard-based dropout module. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.)

6 Results

[30]

The prototype ran 500 replicates per cell across 16 cells with 100 percent convergence and achieved dropout fractions close to targets.

- 8-core `parallel::mclapply` completed 8,000 fits in 421 s with 100 percent convergence.
- Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime; hybrid cells saturate near 1.000 and OL cells near zero.
- Achieved dropout fractions tracked targets: approximately 0.25 for `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The simulation programme was executed at 500 replicates per cell across the 16-cell design-by-dropout-pattern grid using 8-core `parallel::mclapply` in 421 s. Convergence was 100 percent across all 8,000 fits in every cell. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/09-dropout-summary.txt`. Cell MCSE on power runs 0.014 to 0.019 in the binding 0.85 regime, with the hybrid cells effectively saturated near 1.000 and the OL cells near zero. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`.

6.1 Power across designs and dropout patterns

[31]

Power results reproduce the Hendrickson happy-accident ordering: hybrid dominates, OL falls to nominal alpha, with intermediate designs losing power monotonically under heavier dropout.

- Hybrid (approximately 1.00) > OL+BDC (approximately 0.95) > traditional crossover (approximately 0.85) > OL (nominal alpha 0.02–0.05).

- Hybrid dominates because post-week-8 branching produces within-subject and across-path contrasts that survive even severe attrition.
- The ordering reproduces the Hendrickson happy-accident randomisation-path argument at the wider 16-cell grid.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

We begin with the power results. The four designs reproduce the¹ happy-accident randomisation-path ordering at the wider 16-cell grid: the hybrid design (power approximately 1.00) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level near 0.02 to 0.05). The hybrid four-path design dominates because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} that survive even severe attrition. Cell-level power values, percent $\pm$ MCSE:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	2.2 ± 0.7	4.6 ± 0.9	3.4 ± 0.8	4.4 ± 0.9
OL+BDC	95.0 ± 1.0	91.2 ± 1.3	92.4 ± 1.2	89.4 ± 1.4
traditional crossover	85.4 ± 1.6	81.0 ± 1.8	82.4 ± 1.7	76.8 ± 1.9
hybrid	99.8 ± 0.2	100.0 ± 0.0	100.0 ± 0.0	99.4 ± 0.3

[32]

The OL row at nominal alpha is a structural property of Architecture A, not a dropout artefact, and functions as an implicit null check.

- Under Architecture A, D_b is identically one across all OL time-points, so no within-subject $bm:D_{bc}$ contrast exists.
- With no interaction information, the fitter's rejection rates collapse to the nominal type I rate across all dropout cells.
- The OL row is therefore an implicit structural null check, not a power estimate, and should not be interpreted as dropout robustness.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur

sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The OL row sits at nominal alpha under every dropout pattern because Architecture A's level shift on BR at on-drug timepoints is uniform across OL1-OL8: with D_b identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row therefore functions as an implicit null check rather than a power estimate.

[33]

Dropout resilience and absolute power coincide in ranking: hybrid is most robust and most powerful, traditional crossover is least robust and least powerful of the three powered designs.

- Absolute power loss from no-dropout to `biased_40`: hybrid 0.4 pp, OL+BDC 5.6 pp, traditional crossover 8.6 pp.
- The resilience ordering (hybrid > OL+BDC > crossover) coincides with the absolute-power ordering.
- The most powerful design is also the most dropout-robust, a non-trivial alignment with practical design-selection implications.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the `biased_40` cell, runs: hybrid (loss approximately 0.4 percentage points) above OL+BDC (loss approximately 5.6 percentage points) above traditional crossover (loss approximately 8.6 percentage points). Of the three powered designs, the hybrid is most robust and the traditional crossover most vulnerable; it is worth noting that the resilience ordering coincides with the ordering of absolute power, the most powerful design also being the most robust.

6.2 Bias of the biomarker-treatment interaction estimate

[34]

The mean estimated interaction coefficient is stable across dropout cells within each design, indicating no systematic estimand drift.

- $\hat{\beta}_{bm:D_{bc}}$ shows no systematic trend across none, MCAR_25, `biased_25`, and `biased_40` cells within any design.

- Stability holds for the three powered designs (OL+BDC, crossover, hybrid); the OL design is uninformative on bias.
- The absence of estimand drift implies the power degradation traces to standard-error inflation, not bias.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

We turn next to the bias of the biomarker-treatment interaction estimate. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells within each design, indicating no systematic estimand drift under informative dropout in this regime:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	-\$0.09	-\$0.10	-\$0.09	-\$0.10
OL+BDC	-\$2.48	-\$2.48	-\$2.47	-\$2.45
traditional crossover	-\$2.24	-\$2.28	-\$2.25	-\$2.21
hybrid	-\$2.46	-\$2.46	-\$2.46	-\$2.45

[35]

Power degradation under informative dropout traces to standard-error inflation, not estimand drift: empirical SEs expand modestly under biased_40 dropout.

- The largest within-design bias shift (approximately 0.04 for traditional crossover) is within cell-level MCSE.
- Empirical SEs expand under `biased_40`: OL+BDC 0.628 to 0.660, crossover 0.766 to 0.779, hybrid 0.411 to 0.454.
- The power loss in Section 5.1 is therefore attributable to SE inflation rather than systematic bias.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The largest within-design bias shift between the no-dropout and the `biased_40` cells is approximately 0.04 (traditional crossover), well within the cell-level MCSE on the mean (≤ 0.04 for all designs). The power degradation visible in Section 5.1 therefore traces to inflation of the standard error of

1026 $\hat{\beta}_{bm:D_{bc}}$, not to systematic estimand drift. Empirical SEs (last column of
 1027 09-dropout-summary.txt) expand modestly under *biased_40* dropout: from
 1028 0.628 to 0.660 for OL+BDC; from 0.766 to 0.779 for traditional crossover;
 1029 and from 0.411 to 0.454 for hybrid.

1030 6.3 Randomization-path decomposition

1031 [36]

1032 **Path-stratified decomposition of the happy-accident effect is**
 1033 **scoped for a production rerun; the current prototype pro-**
 1034 **vides only marginal results consistent with the hypothesis.**

- 1035 • The current prototype aggregates across paths within each design
 1036 and cannot attribute robustness to specific surviving paths.
- 1037 • The Hendrickson happy-accident hypothesis is consistent with
 1038 the marginal results but requires path-tracked data for a formal
 1039 test.
- 1040 • The follow-up extension requires recording `path_id` per replicate
 1041 and adding path-stratified $\hat{\beta}_{bm:D_{bc}}$ summary logic.

1042 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eius-
 1043 mod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
 1044 minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip
 1045 ex ea commodo consequat. Duis aute irure dolor in reprehenderit in
 1046 voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur
 1047 sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt
 1048 mollit anim id est laborum.

1049 *We note that decomposition of the headline power loss into randomisation-*
 1050 *path-specific contributions is scoped for a production- grade rerun with per-*
 1051 *path-id tracking added to the driver. The current prototype aggregates across*
 1052 *paths within each design and therefore cannot directly attribute the hybrid*
 1053 *design's robustness to specific paths surviving informative dropout. The*
 1054 *substantive¹ 'happy accident' hypothesis (that informative dropout preferen-*
 1055 *tially preserves the most-informative crossover blocks) is consistent with the*
 1056 *present marginal results, but a formal test against the alternative (dropout*
 1057 *uniform across paths) requires the path-tracked data that the current prototype*
 1058 *does not produce. The follow-up driver extension is relatively straightforward*
 1059 *at the simulation level, requiring only that `path_id` be recorded per replicate;*
 1060 *the bottleneck is in the analysis layer, where path-stratified $\hat{\beta}_{bm:D_{bc}}$ distribu-*
 1061 *tions need to be summarised against the cell-pooled estimate.*

1062 6.4 Sensitivity to dropout-pattern misspecification

1063 [37]

1064 **The `biased_25` versus `MCAR_25` contrast isolates the bias mech-**
 1065 **anism from the rate effect; within-design differences are small**
 1066 **and statistically indistinguishable at 500 replicates per cell.**

- OL+BDC: `biased_25` minus `MCAR_25` equals +1.2 pp (CI $[-2.4, +4.8]$); crossover: +1.4 pp (CI $[-3.5, +6.3]$).
- Hybrid is saturated; OL is at null; all differences lie within or near 1.96 cell-MCSE.
- Resolving the distinguishability question would require approximately 2,000 replicates per cell at the current MCSE.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout isolates the bias mechanism from the rate effect. The within-design differences are small in absolute terms: OL+BDC `biased_25` power minus `MCAR_25` power equals +1.2 percentage points (CI $[-2.4, +4.8]$); traditional crossover equals +1.4 percentage points (CI $[-3.5, +6.3]$); hybrid is approximately zero (saturated; bracketed); and OL equals -1.2 percentage points (null cells; bracketed).

[38]

The dominant power determinant is dropout rate rather than bias mechanism; mechanism effects are not statistically distinguishable at 500 replicates per cell.

- Moving from 25% to 40% biased dropout costs 3.0 pp (OL+BDC), 5.6 pp (crossover), 0.6 pp (hybrid).
- The crossover rate-effect loss is clearly detectable above MCSE; the OL+BDC loss is borderline.
- Resolving the `biased_25` versus `MCAR_25` mechanism question requires approximately 2,000 replicates per cell, deferred to a production rerun.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

All four within-design differences lie within 1.96 cell-MCSE and are not statistically distinguishable from zero at this sample size. The dominant effect on power is the dropout rate rather than the bias mechanism: moving from 25 percent to 40 percent biased dropout costs 3.0 percentage points (OL+BDC), 5.6 percentage points (traditional crossover), and 0.6 percentage points (hy-

brid), with the crossover loss clearly detectable above MCSE. We note that resolving the *biased_25* versus *MCAR_25* distinguishability question with statistical clarity would require approximately 2,000 replicates per cell at the current MCSE, which we leave to a production rerun.

7 Design-by-dropout recommendation

[39]

Three classes of anticipated dropout mechanism are mapped to preferred N-of-1 design choices based on the Section 5 simulation results.

- The mapping is indexed by anticipated overall dropout rate within each mechanism class.
- The recommendation draws directly on the power and bias results from the 16-cell prototype simulation.
- The table is intended as practical guidance for trial designers selecting among the four canonical N-of-1 families.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

In this section we shall map three classes of anticipated dropout mechanism (no informative dropout, informative-by-rate, and informative-by-mechanism) to preferred N-of-1 design choices, drawing on the Section 5 results.

Anticipated dropout	Preferred design	Rationale
None or low MCAR ($d \leq 10\%$)	hybrid	Saturated power; no penalty for hybrid's complexity
Moderate MCAR ($d \approx 25\%$)	hybrid	Hybrid retains saturated power; OL+BDC loses 4 pp; crossover loses 4 pp
Heavy MCAR or biased ($d \geq 40\%$)	hybrid	Hybrid at 99% power; OL+BDC drops to 89%; crossover drops to 77%
Worsening-driven (any rate)	hybrid or OL+BDC	OL+BDC's blinded discontinuation phase carries the bias signal explicitly
Improvement-driven (any rate)	OL+BDC	Crossover loses signal when responders drop early; OL+BDC absorbs
Information-poor (OL only)	not recommended	Architecture-A OL has no within-subject $bm:D_{bc}$ contrast

[40]

The dominant recommendation message is that the hybrid design’s power advantage extends to dropout robustness across all three mechanism classes.

- Hybrid is preferred under all dropout rate and mechanism classes except improvement-driven, where OL+BDC is preferred.
- Traditional crossover loses approximately 9 percentage points under `biased_40` and is the least dropout-robust powered design.
- The OL design lacks the within-subject D_{bc} contrast under Architecture A and does not warrant inclusion in the primary recommendation.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The dominant message from the table is that the hybrid Hendrickson design’s power advantage under no dropout extends to robustness against informative dropout. The traditional crossover, despite its strong no-dropout power, loses approximately 9 percentage points under `biased_40` dropout and is the least dropout-robust of the three powered designs. The OL design lacks the within-subject D_{bc} contrast under Architecture A and serves only as a null check at the present prototype’s resolution; it therefore does not warrant inclusion in the primary recommendation.

8 Discussion

8.1 Prototype Monte Carlo confirmation

[41]

The 500-replicate prototype confirms 100 percent convergence across all 8,000 fits with achieved dropout fractions close to targets.

- Executed at $N = 35$, $c_{bm} = 0.45$, Architecture A on 8 cores in 421 s with 100 percent convergence.
- Achieved dropout: approximately 0.25 for `MCAR_25` and `biased_25`; approximately 0.40 for `biased_40`.
- Cell MCSE on power runs 0.014–0.019; hybrid saturates near 1.000.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

A 500-replicate-per-cell prototype run on the full 16-cell design-by-dropout grid (design $\in \{OL, OL+BDC, \text{traditional crossover}, \text{hybrid}\} \times \text{dropout pattern} \in \{\text{none}, \text{MCAR } 25\%, \text{biased } 25\%, \text{biased } 40\%\})$ was executed at $N = 35$, $c_{bm} = 0.45$ under Architecture A using 8-core `parallel::mclapply`. Wall clock was 421 s; convergence was 100 percent across all 8,000 fits. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP summary is at `09-dropout-summary.txt`. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`. Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime, with the hybrid cells effectively saturated near 1.000.

[42]

The prototype reproduces the Hendrickson happy-accident power ordering: hybrid > OL+BDC > crossover > OL at nominal alpha.

- Power ordering: hybrid approximately 1.00, OL+BDC approximately 0.95, crossover approximately 0.85, OL at nominal 0.02–0.05.
- Hybrid dominates because post-week-8 branching produces within-subject and across-path D_{bc} contrasts that survive severe attrition.
- The ordering holds across all four dropout-pattern cells, confirming the Hendrickson happy-accident argument at wider parameter ranges.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The four designs reproduce the¹ happy-accident randomisation-path ordering at the wider grid: hybrid (approximately 1.00 power) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level near 0.02–0.05). The hybrid four-path design dominates because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} that survive even severe attrition.

[43]

The OL design’s nominal-alpha rejection rate is a structural property of Architecture A, not an artefact of dropout, and serves as an implicit null check.

- Under Architecture A, $D_b \equiv 1$ for all OL timepoints, eliminating any within-subject $bm:D_{bc}$ contrast.
- The fitter has no interaction information, so rejection rates collapse to nominal alpha across all dropout conditions.
- This confirms the OL row is a structural null check rather than evidence of dropout robustness.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

The OL row sits at nominal alpha under every dropout pattern because Architecture A’s level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_b identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row therefore functions as an implicit null check, not a power estimate.

[44]

Dropout resilience is dominated by the rate effect, not the bias mechanism; the linear-mixed model with corCAR1 is relatively unbiased under informative dropout in this regime.

- Absolute power loss from no-dropout to `biased_40`: hybrid 0.004, OL+BDC 0.056, crossover 0.086.
- The `biased_25` vs. `MCAR_25` mechanism contrast is within or barely outside 1.96 MCSE; the rate increase from 25 to 40 percent is the dominant effect.
- $\hat{\beta}_{bm:D_{bc}}$ is stable (−2.45 to −2.48), confirming power degradation traces to SE inflation rather than estimand drift.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Resilience to informative dropout, ranked by absolute power loss from the no-

dropout cell to the `biased_40` cell, runs as follows: hybrid (loss approximately 0.004) above OL+BDC (loss approximately 0.056) above traditional crossover (loss approximately 0.086). The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout is small (+0.012, +0.014, -0.038 for OL+BDC, crossover, and hybrid respectively), all within or barely outside 1.96 MCSE. The dominant effect is dropout rate, not bias mechanism: moving from 25 percent to 40 percent biased dropout costs 0.006–0.056 power across designs, while doubling the bias proportion at constant rate is essentially noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells (-2.45 to -2.48 for OL+BDC and hybrid), suggesting that the linear-mixed model with `corCAR1` is relatively unbiased under informative dropout in this regime; the power degradation traces to standard-error inflation rather than systematic estimand drift.

[45]

Two ADEMP caveats limit the prototype’s scope: no formal null cells for non-OL designs, and single-architecture coverage.

- No $c_{bm} = 0$ cells were run for OL+BDC, crossover, or hybrid; production reruns should add explicit null cells for each.
- The prototype is single-architecture (Architecture A); under Architecture B the OL row carries information via the covariance structure.
- Resolving the `biased_25` vs. `MCAR_25` distinguishability question requires approximately 2,000 replicates per cell.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Two ADEMP-discipline caveats warrant mention. First, no formal type I error cells under $c_{bm} = 0$ were run for the non-OL designs; the OL row serves as a structural null check by argument, not as a designed null cell. A canonical production run should add explicit $c_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the prototype is single-architecture (Architecture A) by specification; under Architecture B (MVN) the OL row would carry information because the biomarker-by-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the `biased_25` versus `MCAR_25` distinguishability question would, as noted above, require approximately 2,000 replicates per cell at the current MCSE.

9 Conclusions

[46]

The conclusions distil three findings: design-by-dropout differentiation, rate dominance over mechanism, and SE inflation as the power-loss mechanism.

- The four N-of-1 design families differ materially in dropout robustness in ways the published literature has not characterised.
- Dropout rate dominates mechanism as the determinant of power loss; mechanism becomes load-bearing only when rates are comparable.
- Estimand stability under informative dropout is reassuring but also a warning: coverage degrades as SEs widen without the model acknowledging it.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

In conclusion, three points merit emphasis.

[47]

First conclusion: the hybrid design is the most dropout-robust of the four, losing less than half a percentage point under 40 percent biased dropout at the reference parameters.

- At $N = 35$, $c_{bm} = 0.45$, Architecture A: hybrid loses < 0.5 pp, OL+BDC loses 5.6 pp, crossover loses 8.6 pp under `biased_40`.
- OL lacks the within-subject D_{bc} contrast under Architecture A and is not viable for the biomarker-interaction question.
- The hybrid’s robustness extends the Hendrickson happy-accident argument without requiring path-stratified analysis.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

First, the four candidate aggregated N-of-1 trial designs differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterised. At the prazosin/PTSD reference parameters with $N = 35$, $c_{bm} = 0.45$, and Architecture A, the hybrid Hendrickson design is the most dropout-robust of the four, losing less than half a percentage point of power under 40 percent biased dropout. The OL+BDC de-

sign loses approximately 5.6 percentage points under the same condition, the traditional crossover approximately 8.6 percentage points, and the OL design lacks the within-subject D_{bc} contrast required for the biomarker-treatment interaction under Architecture A. The hybrid's robustness extends and corroborates the¹ happy-accident randomisation-path argument without requiring path-stratified analysis to support the substantive claim.

[48]

Second conclusion: dropout rate is the dominant power determinant; mechanism is load-bearing only when rates are comparable across competing design candidates.

- The `biased_25` vs. `MCAR_25` contrast at fixed 25 percent dropout is below MCSE detection at 500 replicates.
- Moving to 40 percent biased dropout produces clear losses in OL+BDC and crossover; hybrid is unaffected.
- Trial designers should evaluate designs against anticipated dropout rate first, and mechanism only secondarily.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Second, the dominant determinant of the design-by-dropout interaction, in this prototype's evaluation, is the dropout rate rather than the dropout mechanism. The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout produces within-design differences below MCSE detection at 500 replicates per cell; moving to 40 percent biased dropout, by contrast, produces clear power losses in every design except the hybrid. Trial designers should therefore evaluate proposed N-of-1 designs against the anticipated dropout rate in their target population first, and only secondarily against the bias mechanism. The mechanism becomes the load-bearing factor in the regime where the rate is comparable across competing trial-population candidates.

[49]

Third conclusion: estimand stability is reassuring but also a coverage warning, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

- The largest within-design shift in $\hat{\beta}_{bm:D_{bc}}$ is below 0.04 across the no-dropout to `biased_40` range.
- Power degradation traces to SE inflation; the inferential target is preserved under informative dropout in this regime.
- Nominal CI coverage may understate true uncertainty: the coefficient distribution widens but the model does not acknowledge

1390

it.

1391

1392

1393

1394

1395

1396

1397

1398

1399

1400

1401

1402

1403

1404

1405

1406

1407

1408

1409

1410

1411

1412

1413

1414

1415

1416

1417

1418

1419

1420

1421

1422

1423

1424

1425

1426

1427

1428

1429

1430

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Third, the mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells, with the largest within-design shift below 0.04 across the no-dropout to `biased_40` range. The power degradation under informative dropout therefore traces to standard-error inflation rather than systematic estimand drift. This finding is reassuring for the validity of the aggregated N-of-1 framework in the presence of informative dropout, in the sense that the inferential target is preserved, but it is also a warning to trialists that nominal coverage of confidence intervals may understate the true uncertainty in the dropout regime, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

[50]

Three production-grade extensions are scoped: path tracking, Architecture B coverage, and higher replicate counts to resolve the biased-versus-MCAR distinguishability question.

- Per-path-id tracking is needed for the randomisation-path decomposition of the happy-accident effect.
- Architecture B coverage is needed to interpret the OL row as a power estimate rather than a structural null check.
- Resolving the `biased_25` vs. `MCAR_25` question requires 2,000 or more replicates per cell.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Production-grade extensions of the prototype (per-path tracking for the randomisation-path decomposition, Architecture B for the OL design's null-check elevation, and 2,000 or more replicates per cell to resolve the biased-versus-MCAR distinguishability question) are scoped in Section 5 of this manuscript and represent the natural follow-up to the present proof-of-concept evidence.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The informative-dropout module is at `R/censordata.R`.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/09-dropout-replicates.rds`. Morris ADEMP cell-level summaries are at the companion `09-dropout-summary.txt`. Driver scripts are under `analysis/scripts/informative-dropout-by-design/` and `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. The pre-registered ADEMP plan is at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registra`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/09-informative-dropout-by-design/`.

14 References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

- 1462 2. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.
- 1463 3. Little RJA, Rubin DB. *Statistical Analysis with Missing Data*. 3rd ed. Wiley; 2019.
- 1464 4. Molenberghs G, Kenward MG. *Missing Data in Clinical Studies*. Wiley; 2007. doi:10.1002/9780470510445
- 1465 5. O’Kelly M, Ratitch B. *Clinical Trials with Missing Data: A Guide for Practitioners*. Wiley; 2014.
- 1466 6. Mallinckrodt CH, Clark WS, David SR. Accounting for dropout bias using mixed-effects models. *Journal of Biopharmaceutical Statistics*. 2001;11(1-2):9-21. doi:10.1081/BIP-100104194
- 1467 7. Mallinckrodt CH, Clark SWS, Carroll RJ, Molenberghs G. Assessing response profiles from incomplete longitudinal clinical trial data under regulatory considerations. *Journal of Biopharmaceutical Statistics*. 2003;13(2):179-190. doi:10.1081/BIP-120019265
- 1468 8. Prakash A, Risser RC, Mallinckrodt CH. The impact of analytic method on interpretation of outcomes in longitudinal clinical trials. *International Journal of Clinical Practice*. 2008;62(8):1147-1158. doi:10.1111/j.1742-1241.2008.01808.x
- 1469 9. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. ICH E9(R1): Addendum on estimands and sensitivity analysis in clinical trials. Published online 2020.
- 1470 10. Leuchs AK, Zinserling J, Brandt A, Wirtz D, Benda N. Choosing appropriate estimands in clinical trials. *Therapeutic Innovation and Regulatory Science*. 2015;49(4):584-592. doi:10.1177/2168479014567317
- 1471 11. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7
- 1472 12. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 1473 13. Stunnenberg BC, Berends J, Griggs RC, others, Raaphorst J. N-of-1 trials in neurology: A systematic review. *Neurology*. 2022;98(2):e174-e185. doi:10.1212/WNL.0000000000012998

- 1474 14. Natesan Batley P, McClure EB, Brewer B, et al. Evidence and reporting standards in N-of-1 medical studies: A systematic review. *Translational Psychiatry*. 2023;13(1):263. doi:10.1038/s41398-023-02562-8
- 1475 15. Carpenter JR, Kenward MG. *Multiple Imputation and Its Application*. Wiley; 2013.
- 1476 16. Diggle P, Kenward MG. Informative drop-out in longitudinal data analysis. *Journal of the Royal Statistical Society Series C (Applied Statistics)*. 1994;43(1):49-93. doi:10.2307/2986113
- 1477 17. Wulfsohn MS, Tsiatis AA. A joint model for survival and longitudinal data measured with error. *Biometrics*. 1997;53(1):330-339. doi:10.2307/2533118
- 1478 18. Henderson R, Diggle P, Dobson A. Joint modelling of longitudinal measurements and event time data. *Biostatistics*. 2000;1(4):465-480. doi:10.1093/biostatistics/1.4.465
- 1479 19. Verbeke G, Molenberghs G. *Linear Mixed Models for Longitudinal Data*. Springer; 2000.
- 1480 20. Berger VW, Rezvani A, Makarewicz VA. Direct effect on validity of response run-in selection in clinical trials. *Controlled Clinical Trials*. 2003;24(2):156-166. doi:10.1016/s0197-2456(02)00316-1